

In [1]: `!pip install pandas pymongo numpy`

Defaulting to user installation because normal site-packages is not writeable
 Requirement already satisfied: pandas in c:\programdata\anaconda3\lib\site-packages (2.3.3)
 Requirement already satisfied: pymongo in c:\users\dell\appdata\roaming\python\python313\site-packages (4.17.0)
 Requirement already satisfied: numpy in c:\programdata\anaconda3\lib\site-packages (2.3.5)
 Requirement already satisfied: python-dateutil>=2.8.2 in c:\programdata\anaconda3\lib\site-packages (from pandas) (2.9.0.post0)
 Requirement already satisfied: pytz>=2020.1 in c:\programdata\anaconda3\lib\site-packages (from pandas) (2025.2)
 Requirement already satisfied: tzdata>=2022.7 in c:\programdata\anaconda3\lib\site-packages (from pandas) (2025.2)
 Requirement already satisfied: dnspython<3.0.0,>=2.6.1 in c:\users\dell\appdata\roaming\python\python313\site-packages (from pymongo) (2.8.0)
 Requirement already satisfied: six>=1.5 in c:\programdata\anaconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)

In [2]: `import pandas as pd
import numpy as np
from pymongo import MongoClient
from datetime import datetime`

In [3]: `import pandas as pd
import numpy as np
from pymongo import MongoClient
from datetime import datetime

print("Libraries Imported Successfully")`

Libraries Imported Successfully

In [24]: `df = pd.read_csv(
 "individual+household+electric+power+consumption.zip",
 sep=';',
 low_memory=False
)

df.head()`

Out[24]:

	Date	Time	Global_active_power	Global_reactive_power	Voltage	Global_intensi
0	16/12/2006	17:24:00	4.216	0.418	234.840	18.4
1	16/12/2006	17:25:00	5.360	0.436	233.630	23.0
2	16/12/2006	17:26:00	5.374	0.498	233.290	23.0
3	16/12/2006	17:27:00	5.388	0.502	233.740	23.0
4	16/12/2006	17:28:00	3.666	0.528	235.680	15.8



```
In [5]: print(df.shape)
df.info()
```

```
(2075259, 9)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2075259 entries, 0 to 2075258
Data columns (total 9 columns):
 #   Column                Dtype
---  -
 0   Date                  object
 1   Time                  object
 2   Global_active_power   object
 3   Global_reactive_power object
 4   Voltage               object
 5   Global_intensity      object
 6   Sub_metering_1        object
 7   Sub_metering_2        object
 8   Sub_metering_3        float64
dtypes: float64(1), object(8)
memory usage: 142.5+ MB
```

```
In [6]: df.isnull().sum()
```

```
Out[6]: Date                0
Time                0
Global_active_power    0
Global_reactive_power  0
Voltage              0
Global_intensity      0
Sub_metering_1        0
Sub_metering_2        0
Sub_metering_3        25979
dtype: int64
```

```
In [7]: df.replace('?', np.nan, inplace=True)
```

```
In [8]: df['timestamp'] = pd.to_datetime(
    df['Date'] + ' ' + df['Time'],
    format='%d/%m/%Y %H:%M:%S'
)
df[['Date', 'Time', 'timestamp']].head()
```

```
Out[8]:
```

	Date	Time	timestamp
0	16/12/2006	17:24:00	2006-12-16 17:24:00
1	16/12/2006	17:25:00	2006-12-16 17:25:00
2	16/12/2006	17:26:00	2006-12-16 17:26:00
3	16/12/2006	17:27:00	2006-12-16 17:27:00
4	16/12/2006	17:28:00	2006-12-16 17:28:00

```
In [9]: df['Global_active_power'] = pd.to_numeric(
        df['Global_active_power'],
        errors='coerce'
    )
```

```
In [10]: df.dropna(inplace=True)

print(df.shape)
```

(2049280, 10)

```
In [11]: df = df[['timestamp', 'Global_active_power']]

df.head()
```

Out[11]:

	timestamp	Global_active_power
0	2006-12-16 17:24:00	4.216
1	2006-12-16 17:25:00	5.360
2	2006-12-16 17:26:00	5.374
3	2006-12-16 17:27:00	5.388
4	2006-12-16 17:28:00	3.666

```
In [12]: from pymongo import MongoClient

client = MongoClient("mongodb://127.0.0.1:27017/")

db = client["powerdb"]

print("Connected Successfully!")
```

Connected Successfully!

```
In [13]: try:
        db.create_collection(
            "power_data",
            timeseries={
                "timeField": "timestamp",
                "granularity": "minutes"
            }
        )
        print("Collection Created")
    except:
        print("Collection Already Exists")
```

Collection Created

```
In [14]: collection = db["power_data"]
```

```
In [15]: records = df.to_dict("records")

len(records)
```

Out[15]: 2049280

In [16]: `client = MongoClient("mongodb://localhost:27017/")`

In [22]: `db = client['powerconsumptionDB']`

In [24]: `try :
 db.create_collection(
 "power_data",
 timeseries={
 "timefield":"timestamp",
 "granularity":"minutes"
 }
)
 print("collection created")
except Exception as e :
 print(e)`

BSON field 'create.timeseries.timefield' is an unknown field., full error: {'ok': 0.0, 'errmsg': "BSON field 'create.timeseries.timefield' is an unknown field.", 'code': 40415, 'codeName': 'IDLUnknownField'}

In [25]: `collection = db["power_data"]`

In [26]: `records = df.to_dict("records")`

In [27]: `records[3]`

Out[27]: {'timestamp': Timestamp('2006-12-16 17:27:00'), 'Global_active_power': 5.388}

In [2]: `from pymongo import MongoClient

client = MongoClient("mongodb://localhost:27017/")
db = client["PowerConsumptionDB"]`

In [3]: `collection = db["power_data"]`

In [4]: `print(collection)`

Collection(Database(MongoClient(host=['localhost:27017'], document_class=dict, tz_aware=False, connect=True), 'PowerConsumptionDB'), 'power_data')

In [8]: `import pandas as pd

df = pd.read_csv(
 "individual+household+electric+power+consumption.zip",
 sep=";",
 low_memory=False,
 na_values=["?"]
)`

In [13]: `df.head()`

Out[13]:

	Date	Time	Global_active_power	Global_reactive_power	Voltage	Global_intensi
0	16/12/2006	17:24:00	4.216	0.418	234.84	18
1	16/12/2006	17:25:00	5.360	0.436	233.63	23
2	16/12/2006	17:26:00	5.374	0.498	233.29	23
3	16/12/2006	17:27:00	5.388	0.502	233.74	23
4	16/12/2006	17:28:00	3.666	0.528	235.68	18



```
In [14]: df["timestamp"] = pd.to_datetime(
          df["Date"] + " " + df["Time"]
        )
```

C:\Users\DELL\AppData\Local\Temp\ipykernel_1256\1322574899.py:1: UserWarning: Parsing dates in %d/%m/%Y %H:%M:%S format when dayfirst=False (the default) was specified. Pass `dayfirst=True` or specify a format to silence this warning.

```
df["timestamp"] = pd.to_datetime(
```

```
In [11]: df["Global_active_power"] = pd.to_numeric(
          df["Global_active_power"],
          errors="coerce"
        )
```

```
In [12]: df["timestamp"] = pd.to_datetime(
          df["Date"] + " " + df["Time"],
          dayfirst=True
        )
```

```
In [25]: collection.count_documents({})
```

Out[25]: 2049280

```
In [26]: peak = collection.find_one(
          sort=[
              ("Global_active_power",-1)
          ]
        )
print(peak)
```

```
{'_id': ObjectId('6a2bcbb2fb2a20518ea19acf'), 'Date': '22/2/2009', 'Time': '17:09:00', 'Global_active_power': 11.122, 'Global_reactive_power': 0.174, 'Voltage': 229.78, 'Global_intensity': 48.4, 'Sub_metering_1': 35.0, 'Sub_metering_2': 69.0, 'Sub_metering_3': 16.0, 'timestamp': datetime.datetime(2009, 2, 22, 17, 9)}
```

```
In [31]: from datetime import datetime
```

```

start = datetime(2008, 2, 2)
end = datetime(2008, 2, 3)

result = list(
    collection.find(
        {
            "timestamp": {
                "$gte": start,
                "$lt": end
            }
        }
    )
)

print("Total Records:", len(result))

```

Total Records: 1439

```

In [32]: count = collection.count_documents(
    {
        "Global_active_power":{
            "$gt":7
        }
    }
)
print(count)

```

1619

```

In [34]: pipeline = [
    {
        "$group" :{
            "_id":None,
            "Mean":{
                "$avg":
                    "$Global_active_power"
            },
            "Max":{
                "$max":
                    "$Global_active_power"
            },
        }
    }
]

```

```

In [35]: list(
    collection.aggregate(
        pipeline
    )
)

```

Out[35]: [{'_id': None, 'Mean': 1.0916150365006247, 'Max': 11.122}]

```

In [44]: from datetime import datetime

pipeline = [

```

```

{
    "$match": {
        "timestamp": {
            "$gte": datetime(2008, 1, 1),
            "$lt": datetime(2009, 1, 1)
        }
    },
    {
        "$group": {
            "_id": {
                "month": {
                    "$month": "$timestamp"
                }
            },
            "TotalPower": {
                "$sum": "$Global_active_power"
            }
        }
    },
    {
        "$sort": {
            "_id.month": 1
        }
    }
}
]

```

```

In [45]: results = list(collection.aggregate(pipeline))

for r in results:
    print(
        "Month:", r["_id"]["month"],
        "Total Power:", round(r["TotalPower"], 2)
    )

```

```

Month: 1 Total Power: 65169.38
Month: 2 Total Power: 49331.07
Month: 3 Total Power: 55590.59
Month: 4 Total Power: 48209.99
Month: 5 Total Power: 45721.86
Month: 6 Total Power: 42943.97
Month: 7 Total Power: 35477.42
Month: 8 Total Power: 12341.88
Month: 9 Total Power: 42667.79
Month: 10 Total Power: 50696.46
Month: 11 Total Power: 59908.76
Month: 12 Total Power: 56833.91

```

```

In [49]: import matplotlib.pyplot as plt

months = [
    r["_id"]["month"]
    for r in results
]

power = [

```

```
r["TotalPower"]
for r in results
]

plt.figure(
    figsize=(10,5)
)

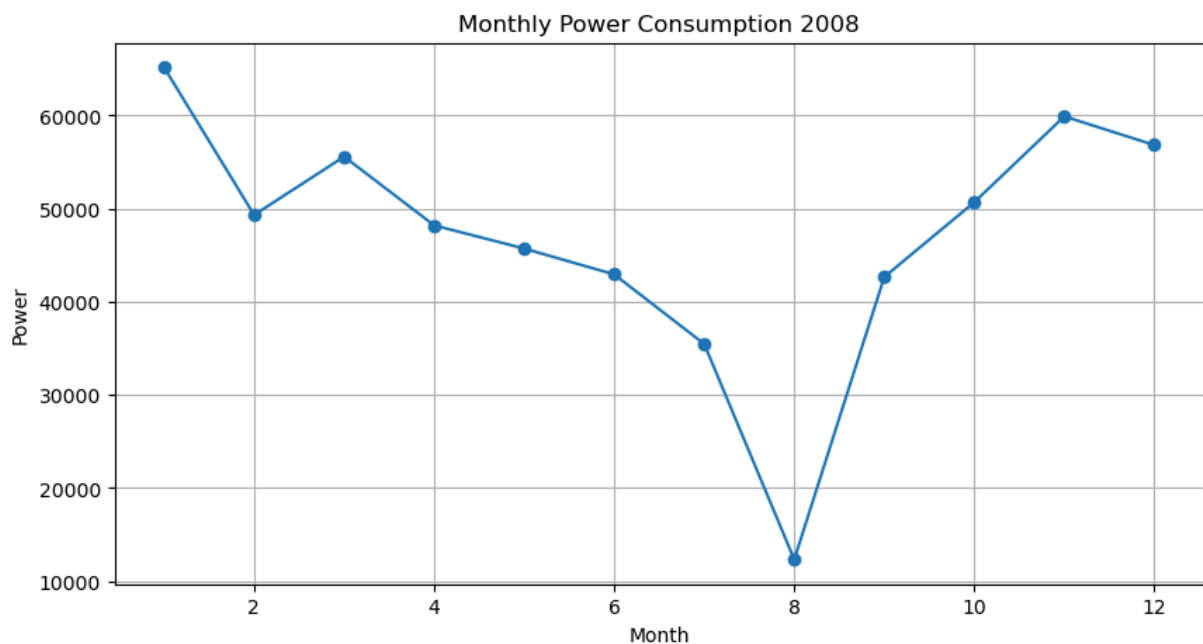
plt.plot(
    months,
    power,
    marker='o'
)

plt.title(
    "Monthly Power Consumption 2008"
)

plt.xlabel("Month")
plt.ylabel("Power")

plt.grid(True)

plt.show()
```



In []: